



PETE 451 Chemistry of Geothermal Systems

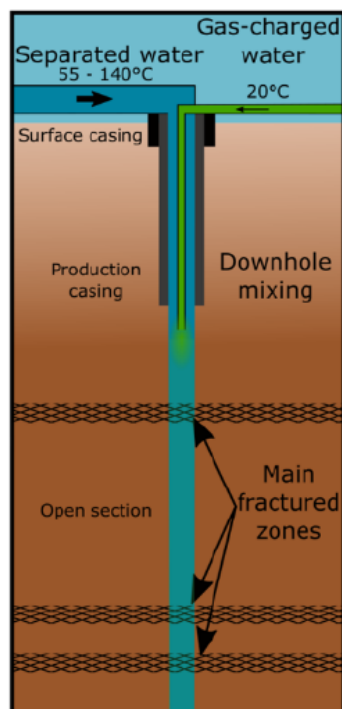
Advective Reactive transport 1D with Phreeqc

Part-8.2

1D Advective Reactive Transport

An example here for 1D advective reactive transport process mimicking a flow path (a fracture zone) along an injection wellbore.

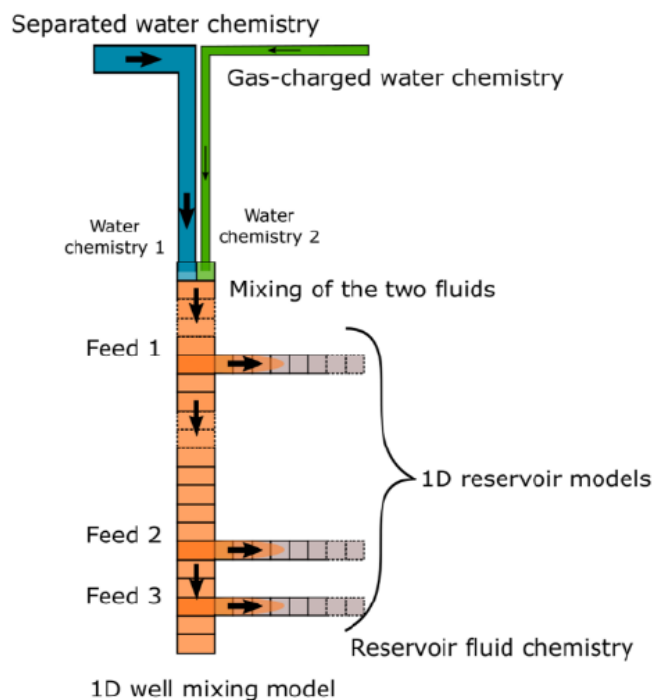
CarbFix injection system
HN-16



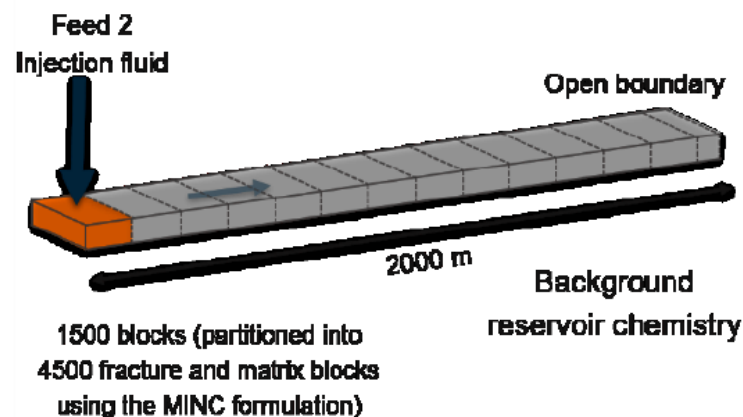
Elevation (masl)

400
200
0
-200
-400
-600
-800
-1000
-1200
-1400
-1600

Reservoir models



1D reservoir model at feed 2

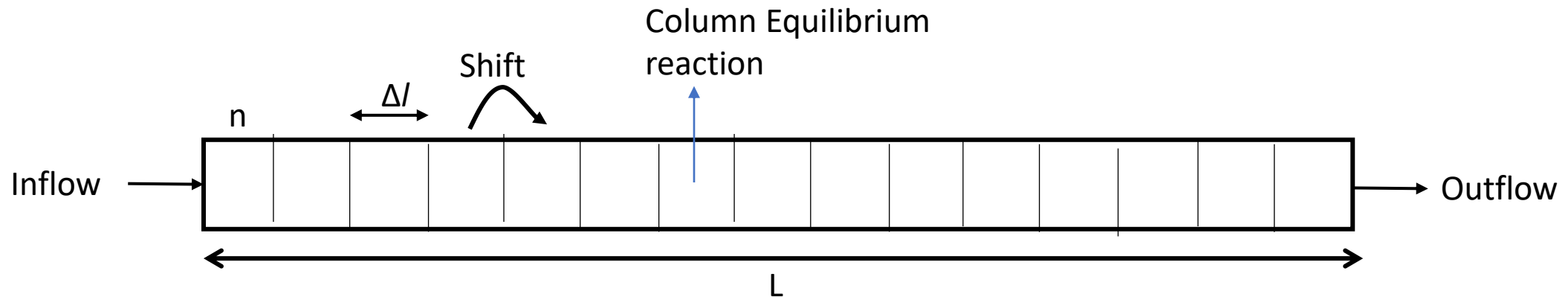


Water chemistry	
	Separated water
	Gas-charged condensate
	CarbFix fluid = Mixture of separated water and gas charged condensate
	Reservoir fluid

Ratouis et al. (2021)

1D Advective Reactive Transport

1D advective reactive transport in Phreeqc



Every time step the inflow (defined as solution 0) shifts to one cell and during each time step the fluid and the minerals are equilibrated and transported to next cell

Inflow (Solution 0)

Initial column solution (solution 1 – n) transport

Cells = n

Column length = L

Cell length = $L / n = \Delta l$

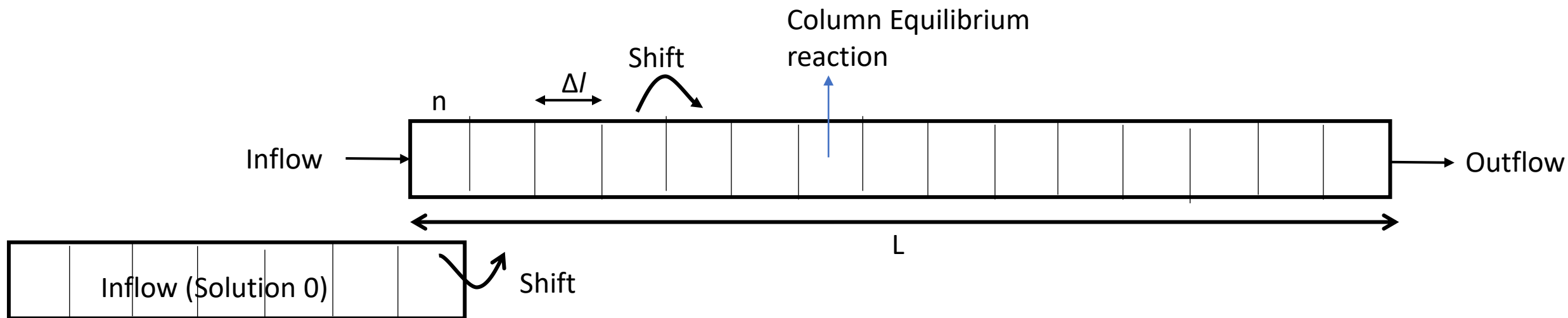
Time step = Δt

Flow velocity = $u = \Delta l / \Delta t$ (m/s)

Shifts

1D Advective Reactive Transport

1D advective reactive transport in Phreeqc



Every time step the inflow (defined as solution 0) shifts to one cell and during each time step the fluid and the minerals are equilibrated and transported to next cell

Inflow (Solution 0)

Initial column solution (solution 1 – n) transport

Cells = n ; Column length = L ; Cell length = $L / n = \Delta l$; Time step = Δt ; Flow velocity = $u = \Delta l / \Delta t$ (m/s); Shifts

1D Advective Reactive Transport

The advective-dispersive transport capabilities of PHREEQC are derived from a formulation of 1D, advective-dispersive transport presented by Appelo and Postma (2005).

The 1D column is defined by a series of cells, each of which has the same pore volume.

Lengths are defined for each cell and the time step (time step) gives the time necessary for a pore volume of water to move through each cell. Thus, the velocity of water in each cell is determined by the length of the cell divided by the time step.

In an example, a column of XX cells is modeled and XX pore volumes of filling solution are moved through the column (number of shifts/number of cells).

At each shift, advection is simulated by moving solution cells to cell, until solution 0 is moved to endcell (upwind scheme). With flux-type boundary conditions, the dispersion steps follow the advective shift.

1D Advective Reactive Transport

With Dirichlet boundary conditions, the dispersion step and the advective shift are alternated.

After each advective shift and dispersion step, kinetic reactions and chemical equilibria are calculated.

The moles of pure phases and the compositions of the exchange assemblage, surface assemblage, gas phase, solid-solution assemblage, and kinetic reactants in each cell are updated after each chemical calculation.

For advective transport, the influent solution must be defined, otherwise, the program stops with an error message. Solution 0 is the influent with *-flow_direction forward*; solution cells + 1 is the influent when the flow is backward. If the effluent (solution cells + 1, for a direction forward) is not defined, the program copies it from the effluent boundary cell in the column.

1D Advective Reactive Transport

A closed boundary condition is not possible with the advective flow, and the boundary condition will be changed to a flux boundary condition.

Likewise, a flux-boundary condition is not possible when pure diffusion is modeled, and the boundary condition will be changed to a closed boundary condition.

The *-time_step* identifier defines the length of time associated with each advective shift or diffusion period.

The program may subdivide this time step into smaller dispersion time steps if necessary to calculate dispersion accurately. Each dispersion time step may be further subdivided to integrate the kinetic reactions (KINETICS data block). Kinetic reactions are likely to slow the calculations by a factor of six or more compared to pure equilibrium calculations.

1D Advective Reactive Transport

The numerical scheme is for cell-centered concentrations, which has consequences for data interpretation. Thus, the composition in a boundary cell is a half-cell distance away from the column outlet and needs a half time step to arrive at (or from) the column end.

A “*dual porosity*” model, in which part of the porosity allows advective flow and part of the porosity is accessible only by diffusion, can be developed with a first-order exchange model or with finite-differences, and both approaches can be defined in terms of a mixing among cells.

With the TRANSPORT data block, one column of mobile cells is used to represent the part of the flow system in which advection occurs, and then additional immobile cells connected to the mobile cells are used to represent the stagnant zone that is accessible only by diffusion.

The stagnant zone can be defined to be parallel or perpendicular to the column of mobile cells or to be a combination of the two by proper definition of mixing factors in MIX data blocks.

A shortcut is available for the classical formulation of a dual porosity medium with a first-order rate of exchange. In this case, *-stagnant* is used to define one stagnant cell for each mobile cell (`stagnant_cells = 1`), an exchange factor (`exchange_factor`) for the exchange between immobile and mobile cells, and the porosities and for the mobile and immobile cells.



References

Ratouis, T.M.P., Snæbjornsdottir, S.O., Sigfússon, B., Gunnarsson, I., Voigt, M.J., Aradottir, E.S., 2020. Reactive Transport Model of CO₂ and H₂S Mineral Sequestration At the CarbFix2 Reinjection site, Hellisheiði Geothermal Power Plant. World Geothermal Congress, Iceland. SW-Iceland.

Appelo, C.A.J.; Postma, D. (2005) Geochemistry, Groundwater and Pollution: Second Edition. Amsterdam, the Netherlands.